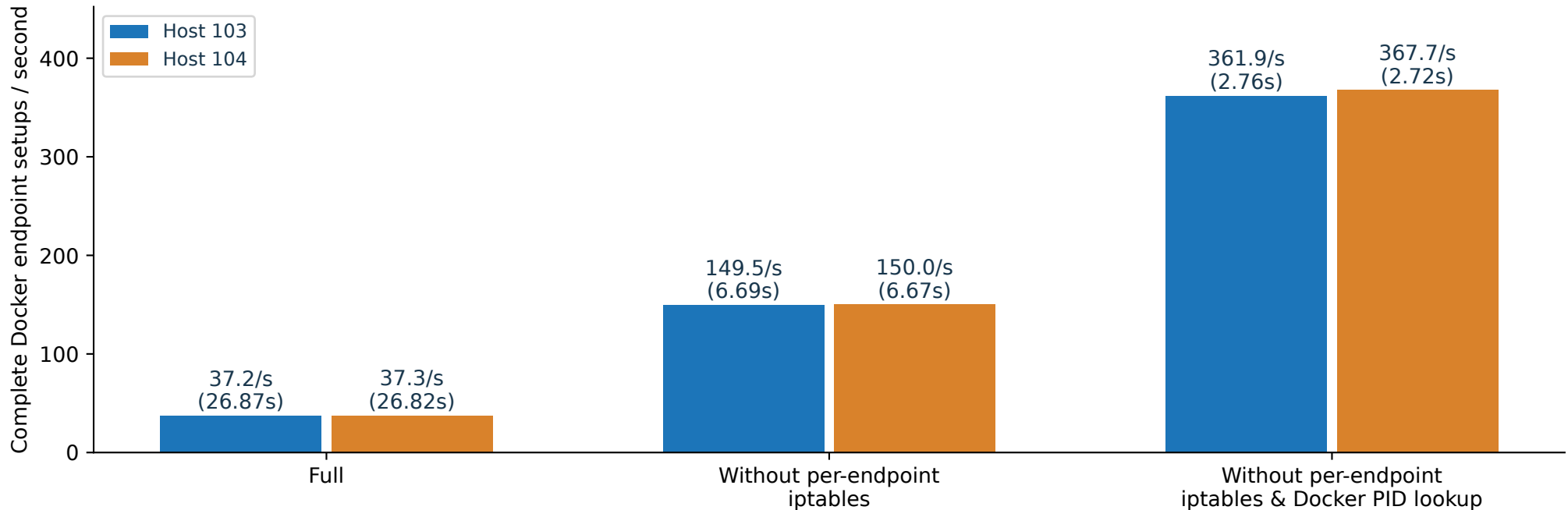


Full bare Docker setup: three measured cases

1,000 complete Docker-backed endpoints per case and host • 12 running containers • 32 workers / eight CLI slots



Full: 1,000 iptables policy updates and 1,000 Docker PID lookups.

Without iptables: policy prepared once; all 1,000 Docker PID lookups remain.

Without iptables & Docker PID: policy and the 12 container PIDs prepared before timing.

Every case still performs 1,000 sysctl updates, creates 1,000 unique VXLANs/VNIs/UDP ports and bridges, attaches 1,000 veth peers to actual Docker namespaces, and installs 2,000 IPsec states plus 2,000 policies. All six runs succeeded with zero errors. Container PIDs/start times were unchanged, and cleanup left zero test veths in containers.

These tests exercise the complete Docker-backed setup branch. Container namespace reuse replaces standalone namespace creation; Docker mode does not add a default route. The earlier 234-251/s result was for standalone endpoints and must not be labelled Docker throughput. All setup-operation rates and exact commands follow on pages 2-5.

Bare provisioning only; excludes control-plane work and traffic. Product forwarding and PID caching are unchanged.

Every step: exact operation and bare throughput

Part 1/3 • Each rate counts completed endpoint steps/s; grouped operations are marked • 1,000 items

STEP / API	COMMAND OR NETLINK ATTRIBUTES	103 /s	104 /s
Docker PID lookup (container alternative) CLI / Docker API	<code>docker inspect -f '{{.State.Pid}}' CONTAINER</code>	344	338
Namespace creation CLI / unshare + mount	<code>ip netns add NS</code>	1,591	1,407
Stale veth lookup RTM_GETLINK	<code>IFLA_IFNAME=VETH_OUT; single-name lookup, no NLM_F_DUMP</code>	12,260	14,113
Veth pair creation into namespace RTM_NEWLINK	<code>kind=veth; name=VETH_OUT; mtu=1080; group=GROUP; peer={name=VETH_IN, mtu=1080, NET_NS_PID=PID (Docker) or NET_NS_FD=NS_FD (standalone)}</code>	1,487	1,455
Namespace address CLI / RTM_NEWADDR	<code>nsenter -t PID -n ip addr add NS_IP/30 dev VETH_IN (Docker); standalone: nsenter --net=/var/run/netns/NS -- ip addr add NS_IP/30 dev VETH_IN</code>	2,666	2,382
Namespace interface UP CLI / link update	<code>nsenter -t PID -n ip link set VETH_IN up (Docker); standalone: nsenter --net=/var/run/netns/NS -- ip link set VETH_IN up</code>	2,255	2,198
Namespace default route CLI / RTM_NEWROUTE	<code>nsenter --net=/var/run/netns/NS -- ip route add default via BRIDGE_IP</code>	2,449	2,601
Stale bridge lookup RTM_GETLINK	<code>IFLA_IFNAME=BRIDGE; single-name lookup, no NLM_F_DUMP</code>	10,842	13,573

Names/IPs are symbolic: the harness substitutes isolated lab values. Individual namespace/veth rates use standalone namespaces; full tests on page 1 use Docker PIDs and existing container namespaces. Native link updates use RTM_SETLINK, matching Nullnet. Stale lookups found no existing device.

Every step: exact operation and bare throughput

Part 2/3 • Each rate counts completed endpoint steps/s; grouped operations are marked • 1,000 items

STEP / API	COMMAND OR NETLINK ATTRIBUTES	103 /s	104 /s
Bridge creation RTM_NEWLINK	kind=bridge; name=BRIDGE; group=GROUP	3,974	4,473
Bridge and veth lookup 2 x RTM_GETLINK	IFLA_IFNAME=BRIDGE, then IFLA_IFNAME=VETH_OUT (two lookups per endpoint)	8,421	8,087
Bridge address RTM_NEWADDR	family=AF_INET; prefixlen=30; index=BRIDGE_IFINDEX; IFA_ADDRESS=IFA_LOCAL=BRIDGE_IP	12,434	13,049
Bridge MTU and UP RTM_SETLINK	index=BRIDGE_IFINDEX; IFLA_MTU=1080; IFF_UP flag and change bit	4,852	4,750
Veth bridge attach MTU UP RTM_SETLINK	index=VETH_OUT_IFINDEX; IFLA_MASTER=BRIDGE_IFINDEX; IFLA_MTU=1080; IFF_UP	1,679	1,790
Four stale cross-host lookups 4 x RTM_GETLINK	IFLA_IFNAME in {macsec-ID-s, macsec-ID-c, veth-ID-s, vxlan-NS}; four single-name lookups per endpoint	4,332	4,141
VXLAN creation RTM_NEWLINK	kind=vxlan; name=vxlan-NS; group=GROUP; IFLA_VXLAN_ID=VNI; LOCAL=LOCAL_IP; GROUP=REMOTE_IP; PORT=UDP_PORT	4,677	4,623
VXLAN lookup RTM_GETLINK	IFLA_IFNAME=vxlan-NS; single-name lookup, no NLM_F_DUMP	12,030	12,999

Names/IPs are symbolic: the harness substitutes isolated lab values. Individual namespace/veth rates use standalone namespaces; full tests on page 1 use Docker PIDs and existing container namespaces. Native link updates use RTM_SETLINK, matching Nullnet. Stale lookups found no existing device.

Every step: exact operation and bare throughput

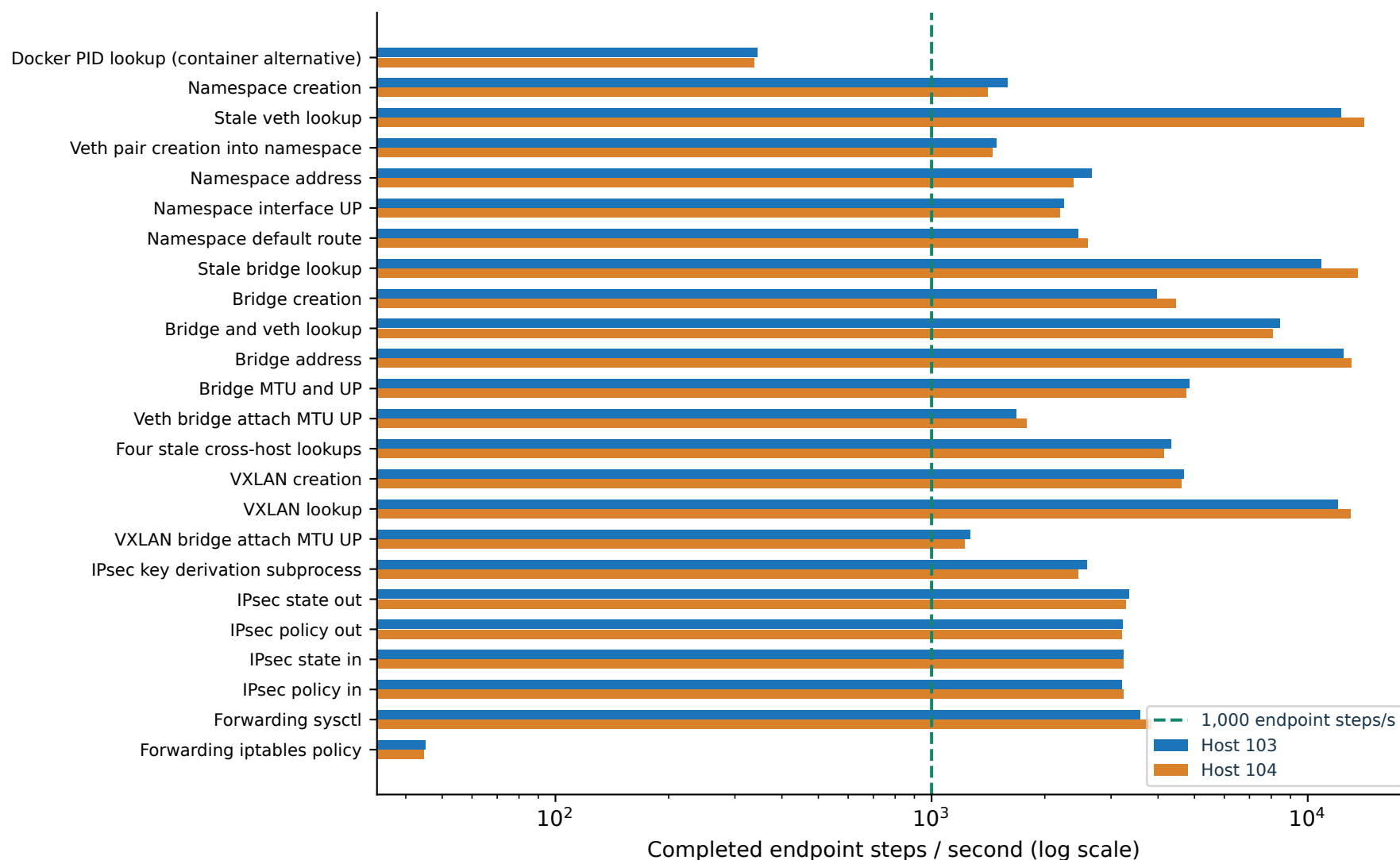
Part 3/3 • Each rate counts completed endpoint steps/s; grouped operations are marked • 1,000 items

STEP / API	COMMAND OR NETLINK ATTRIBUTES	103 /s	104 /s
VXLAN bridge attach MTU UP RTM_SETLINK	index=VXLAN_IFINDEX; IFLA_MASTER=BRIDGE_IFINDEX; IFLA_MTU=1080; IFF_UP	1,266	1,223
IPsec key derivation subprocess CLI	sha256sum (stdin: ASCII KEY_HEX); take first 8 digest hex characters as salt; AEAD_KEY=0x + KEY_HEX + salt	2,594	2,454
IPsec state out CLI / NETLINK_XFRM	ip xfrm state add src LOCAL_IP dst REMOTE_IP proto esp spi SPI aead 'rfc4106(gcm(aes))' AEAD_KEY 128 mode transport	3,355	3,280
IPsec policy out CLI / NETLINK_XFRM	ip xfrm policy add src LOCAL_IP dst REMOTE_IP proto udp dport UDP_PORT dir out tmpl src LOCAL_IP dst REMOTE_IP proto esp spi SPI mode transport	3,222	3,215
IPsec state in CLI / NETLINK_XFRM	ip xfrm state add src REMOTE_IP dst LOCAL_IP proto esp spi SPI aead 'rfc4106(gcm(aes))' AEAD_KEY 128 mode transport	3,241	3,233
IPsec policy in CLI / NETLINK_XFRM	ip xfrm policy add src REMOTE_IP dst LOCAL_IP proto udp dport UDP_PORT dir in tmpl src REMOTE_IP dst LOCAL_IP proto esp spi SPI mode transport	3,203	3,246
Forwarding sysctl CLI / procfs	sysctl -w net.ipv4.ip_forward=1	3,585	3,839
Forwarding iptables policy CLI / nftables backend	iptables -P FORWARD ACCEPT	45	45

Names/IPs are symbolic: the harness substitutes isolated lab values. Individual namespace/veth rates use standalone namespaces; full tests on page 1 use Docker PIDs and existing container namespaces. Native link updates use RTM_SETLINK, matching Nullnet. Stale lookups found no existing device.

Individual operations are fast; complete setup adds their costs

Final independent-stage measurements • 32 workers • At most eight simultaneous CLI commands



Prerequisites for each stage were prepared before its timer. The full Docker benchmark on page 1 includes every Docker-backed operation. Only the named operations are moved before timing; sysctl remains per endpoint in all three cases. Standalone namespace creation/default route are measured separately as alternative-path operations.

Achieved harness rates, not kernel maxima. All errors zero; test resources removed; product source unchanged. All six Docker full-setup runs were untraced.